Predicting Customer Churn at SyriaTel

Using Machine Learning for Strategic Retention

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Business Understanding

Key Stakeholder

SyrialTel marketing teams

Strategic Objectives

SyriaTel is experiencing customer churn, leading to **revenue loss** and increased costs associated with **acquiring new customers**.

The goal is to develop a predictive model that identifies at-risk customers early.

Targeted marketing

Improved customer service

Personalized retention strategies

A key focus is **minimizing missed churners**, as failing to identify these customers represents **lost revenue**.

Data Understanding

Data Source

SyrialTel Telecommunications

Historical Data

3,333

Dataset Profile

Records with zero missing values

Features

Behavioural and service-related variables

Predictive

Indicators ready for churn modeling

Data Preparation

1

Feature Engineering

Removed irrelevant features such as phone number and state to focus on predictive signals.

2

Encoding & Splitting

Encoded categorical variables like international plan and voicemail plan; split data into training and testing sets.

3

Scaling & Integrity

Applied feature scaling for Logistic Regression and prevented data leakage by fitting transformations only on training data.

Modeling Approach

PHASE 1

Baseline

Logistic Regression

- Simple and interpretable
- Provided a performance benchmark
- High accuracy but poor recall for churners

PHASE 2

Improved

Decision Tree

- Captures non-linear relationships
- Significantly improved recall
- Better balance between precision and recall

PHASE 3

Refined

Tuned Decision Tree

- Hyperparameters adjusted to control complexity
- Aimed to reduce overfitting
- No significant recall gain over base model

Logistic Regression

Benchmark Performance

This model established the **benchmark** and why it was useful for comparison.

> It emphasizes that although the baseline achieved **high accuracy**, it performed **poorly at identifying churners**.

BASELINE RECALL

0.18

⚠️ Poor Churn Identification

IMPROVED RECALL

0.73

Significant Detection Gain

Decision Tree

Capturing Complexity

This model improved the **churn detection objective**.

It highlights the **Decision Tree's ability** to capture **non-linear relationships** and provide a better balance between **precision and recall**.

Tuned Decision Tree

Complexity Management

Refinement effort used to **manage model complexity** and **reduce overfitting**.

> It makes clear that tuning did not meaningfully improve **recall** beyond the base Decision Tree.

Model Stability

Overfitting Protection Achieved

0.92 Accuracy Maintained

Evaluation

Accuracy

Precision

RECALL

PRIMARY METRIC

F1-score

Why Recall Matters:

Missing a churner means **losing a customer** without intervention, which **directly impacts revenue**.

Model Comparison

Performance Matrix

Model	Accuracy	Precision	Recall	F1Score
Logistic Regression	0.86	0.60	0.18	0.27
Decision Tree	0.92	0.73	0.73	0.73
Tuned Decision Tree	0.92	0.73	0.71	0.72

Conclusion

The **Decision Tree model** provides the **best balance** between accuracy and recall, making it the **most suitable model** for SyriaTel's business problem.

Business Recommendations

Risk Identification

Use the Decision Tree model to identify customers at high risk of churn before they leave the network.

Retention Tactics

- Personalized offers
- Proactive customer support
- Service or pricing adjustments

Revenue Protection

Focus on early identification to reduce customer attrition and protect corporate revenue.

Limitations

01

Constraints

Based on historical data; may not capture future behavior changes.

02

External Factors

External factors like competitor pricing are not currently included.

03

Class Imbalance

Slight class imbalance may affect specific model performance segments.

Next Steps

Advanced Modeling

Explore Random Forest and Gradient Boosting to further refine prediction accuracy.

Data Enrichment

Perform feature engineering and incorporate additional data sources for a 360-degree view.

Operationalization

Deploy model for real-time churn monitoring and continuously retrain as behavior evolves.


ANY QUESTIONS?
THANK YOU!